Exp 7 FLOW CONTROL USING SLIDING CONTROL PROTOCOL

Aim:

To write a program to implement flow control at data link layer using SLIDING WINDOW PROTOCOL.

SENDER:

* Get the window size from user
* Get the txt message
* Divide the messages into frames
* Send to SENDER-BUFFER
* Wait for Acknowledgement
* Read RECEIVER_BUFFER
* If Ack is recieved, send the next
* If NACK is recieved, retransmit the frame.

RECEIVER:

* Read frames from SENDER_BUFFER.
* Check the frame number
* If the frame number is expected send Ack.
* otherwise send NACK.
* write ACK/NACK into RECEIVER-BUFFER.

CODE:

SENDER

```
import time
w = int(input("window size:"))
msg = input("Message:")
i = 0
while i < len(msg):
    end = min(i+w, len(msg))
    with open("sender_BUFFER", "w") as f:
        for j in range(i, end):
            f.write(f"{j}:{msg[j]}/n")
    print("Frame", "j", "→", msg[j])
    time.sleep(2)
    With open("Reciever_BUFFER") as f:
        ack = f.read()
    i = int(ack[1])
print("Transmission completed")
```

RECEIVER

```
import time
import os
import excepted=0
while true:
    if not os.path.exists("sender_Buffer")
        Continue
    with open("sender_Buffer") as f:
        frames = f.readline()
```

```
n = int(x.split(":")[0])
if n == expected:
    expected += 1
    print("Recieved frame", n);
    with open("Recieved_Buffer","w")
    as f:
        f.write("ACK"+str(excepted))
else:
    with open("Received_Buffer","w")
                as f:
    f.write("NACK"+str(expected))
    break
time.sleep(1)
```

Result:

The SLIDING WINDOW PROTOCOL was implemented successfully. ACK/NACK was simulated successfully.